



Dynamic networks of methicillin-resistant *Staphylococcus aureus* in communities drive hospital transmission as revealed by whole-genome sequencing



Scott A. Cohen^{1,2}, Massimiliano S. Tagliamonte^{1,3}, Carla Mavian^{1,3}, Mattia Prosperi², Marco Salemi^{1,3}, J. Adrian Tyndall, J. Glenn Morris¹

¹Emerging Pathogens Institute; ²Department of Epidemiology, College of Medicine & PHHP; ³Department of Pathology, Immunology, Laboratory Medicine, College of Medicine.

INTRODUCTION

BACKGROUND

- Methicillin-resistant *Staphylococcus aureus* (MRSA) infections remain a leading cause of antibiotic resistant infections in hospitals
- Recent MRSA trends have promoted uncertainty on transmission pathways, particularly among strains in hospital and community settings.
- A simultaneous application of whole-genome sequencing of both community and hospital *S. aureus* isolates will elucidate modern evolutionary pathways.

OBJECTIVE:

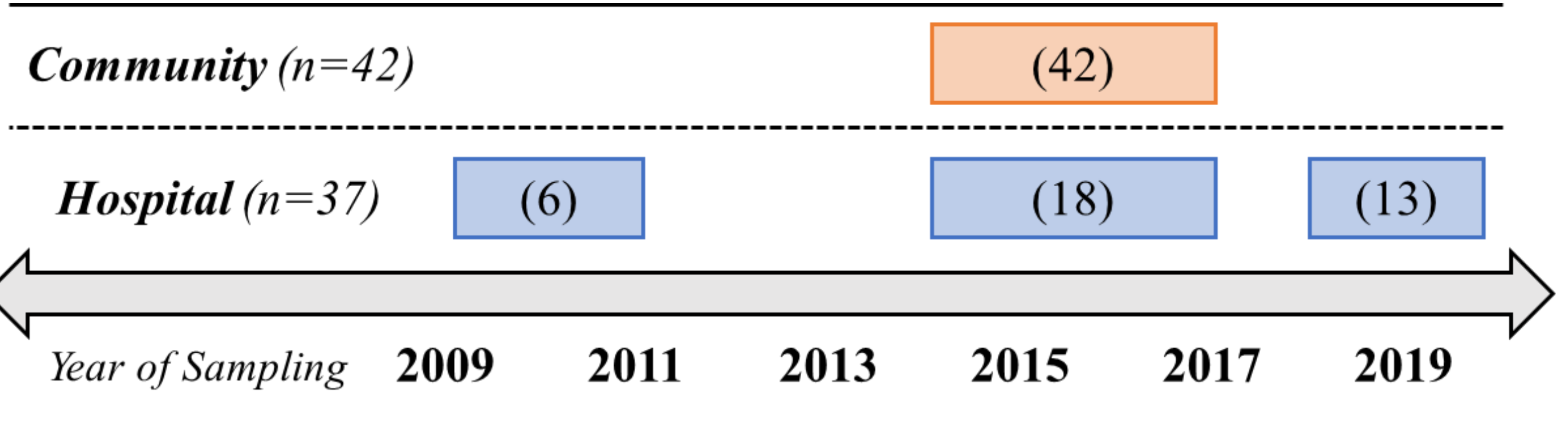
- To elucidate MRSA transmission networks within and between hospital and community settings, we performed longitudinal epidemiological and phylogenetic analyses on MRSA infections in North-Central Florida.

METHODS

STUDY DESIGN

- Serial cross-sectional community and hospital sampling were performed at University of Florida Health between 2010 and 2019 (Figure 1)
- Community Sampling:** From August 2015-January 2017, pediatric and adult patients presenting to ED with acute skin and soft tissue infection (SSTI) were prospectively enrolled and surveyed (Figure 2)
- Hospital Sampling:** MRSA isolates collected from healthcare-acquired infections were sampled before (2010), concurrently, and after (2019) community sampling study period. (Figure 2)
- Whole-genome sequencing (Illumina) was conducted on cultured isolates

Figure 1: Study timeline of community and hospital MRSA isolate sampling



STATISTICAL APPROACH

- Genotype of isolates were characterized via spa-typing (Figure 3)
- Bayesian phylodynamic inference framework was implemented to assess the transmission network between hospital and community settings (Figure 4)
- Fisher's Exact test and Wilcoxon rank-sum were used to test associations between risk factors with microbiological among community isolates. (Table 1)
- Kernel density models were developed to illustrate the spatial distribution of the community-based MRSA and MSSA SSTI incidence (Figure 5)

RESULTS OVERVIEW: TABLES & FIGURES

Figure 2: Flowchart of study inclusion

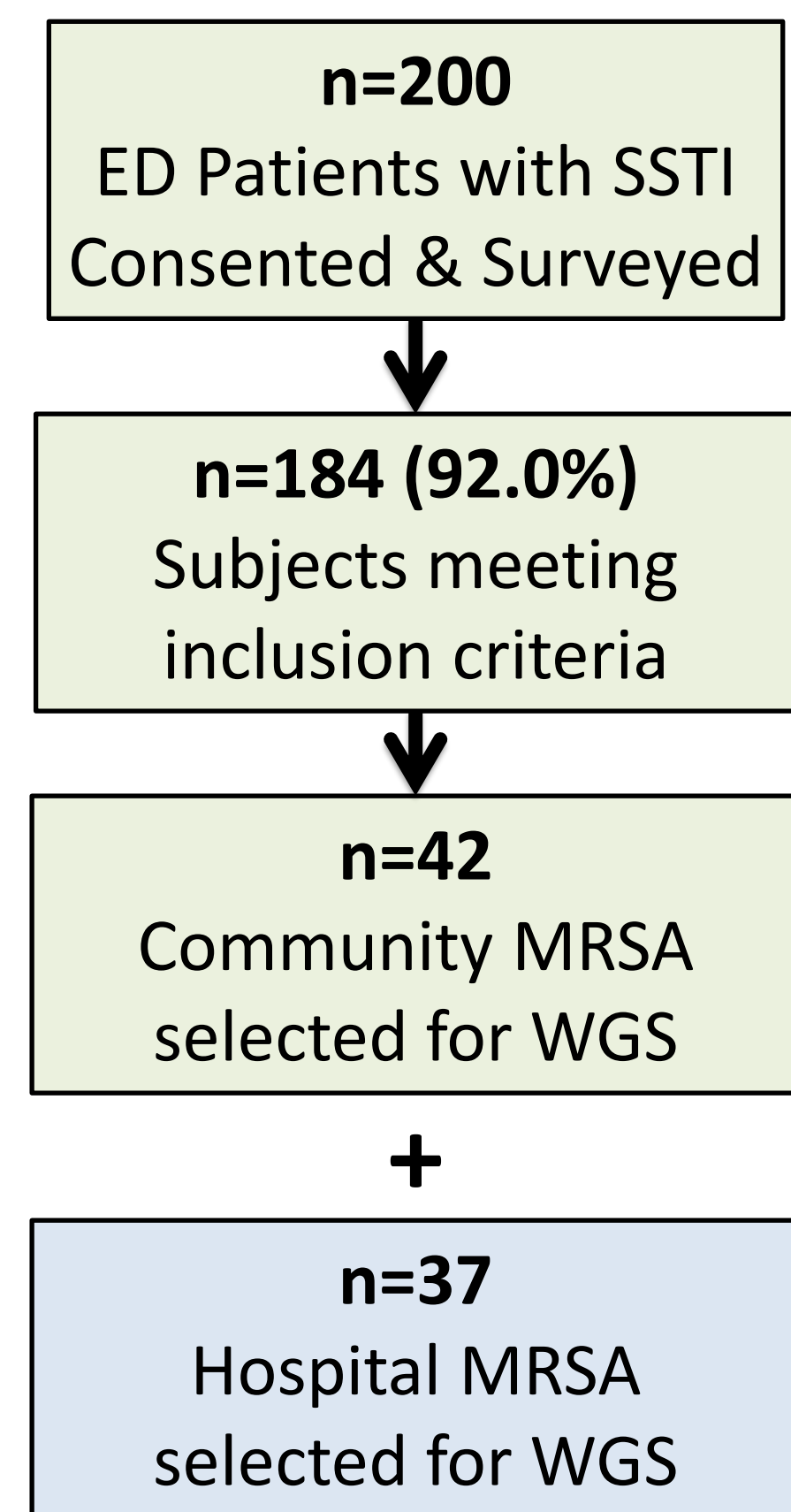


Figure 3: Relative proportions of spa types of MRSA isolates by sampling location and year

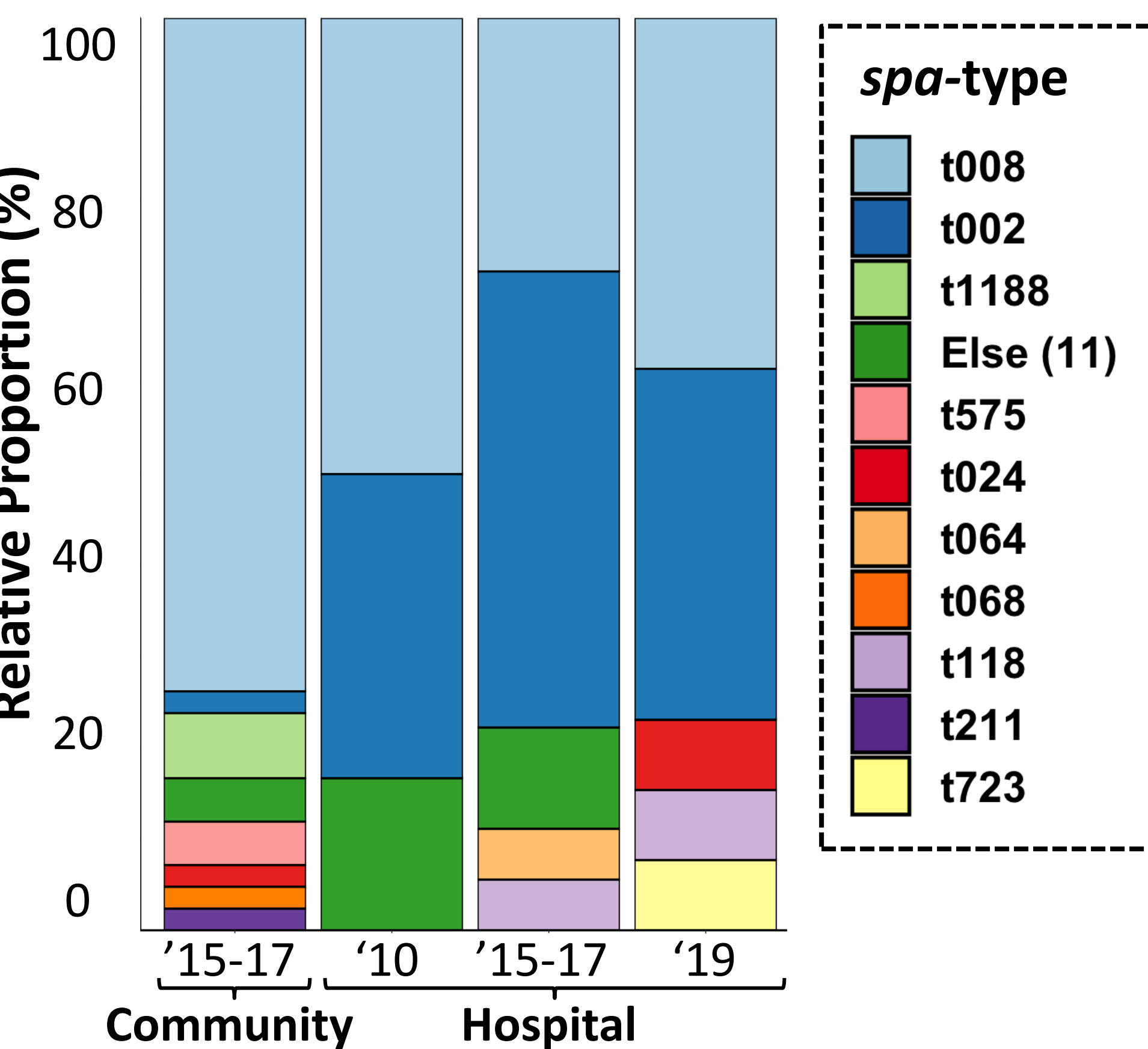


Figure 4: Bayesian maximum clade credibility tree of MRSA by major clade

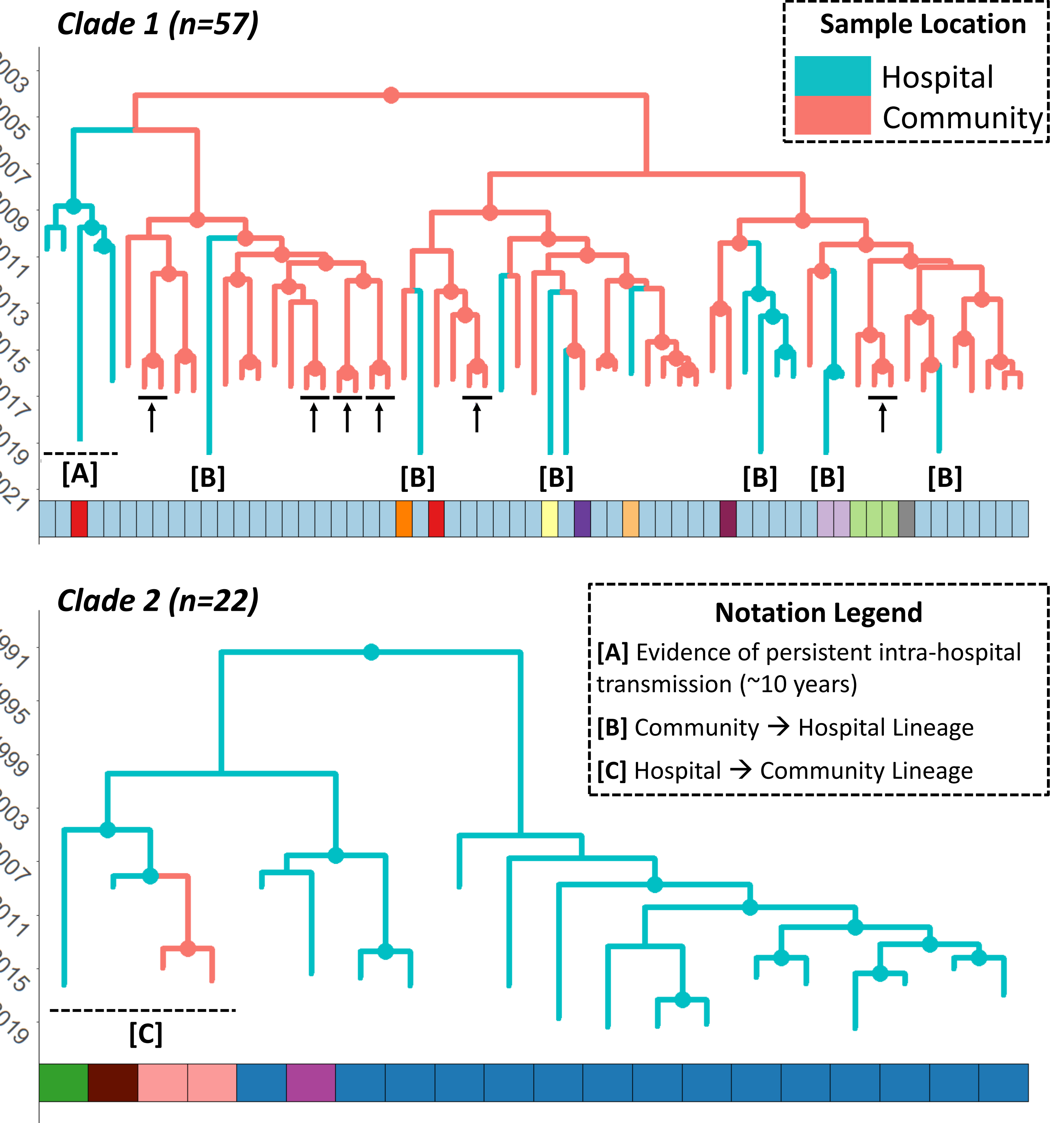
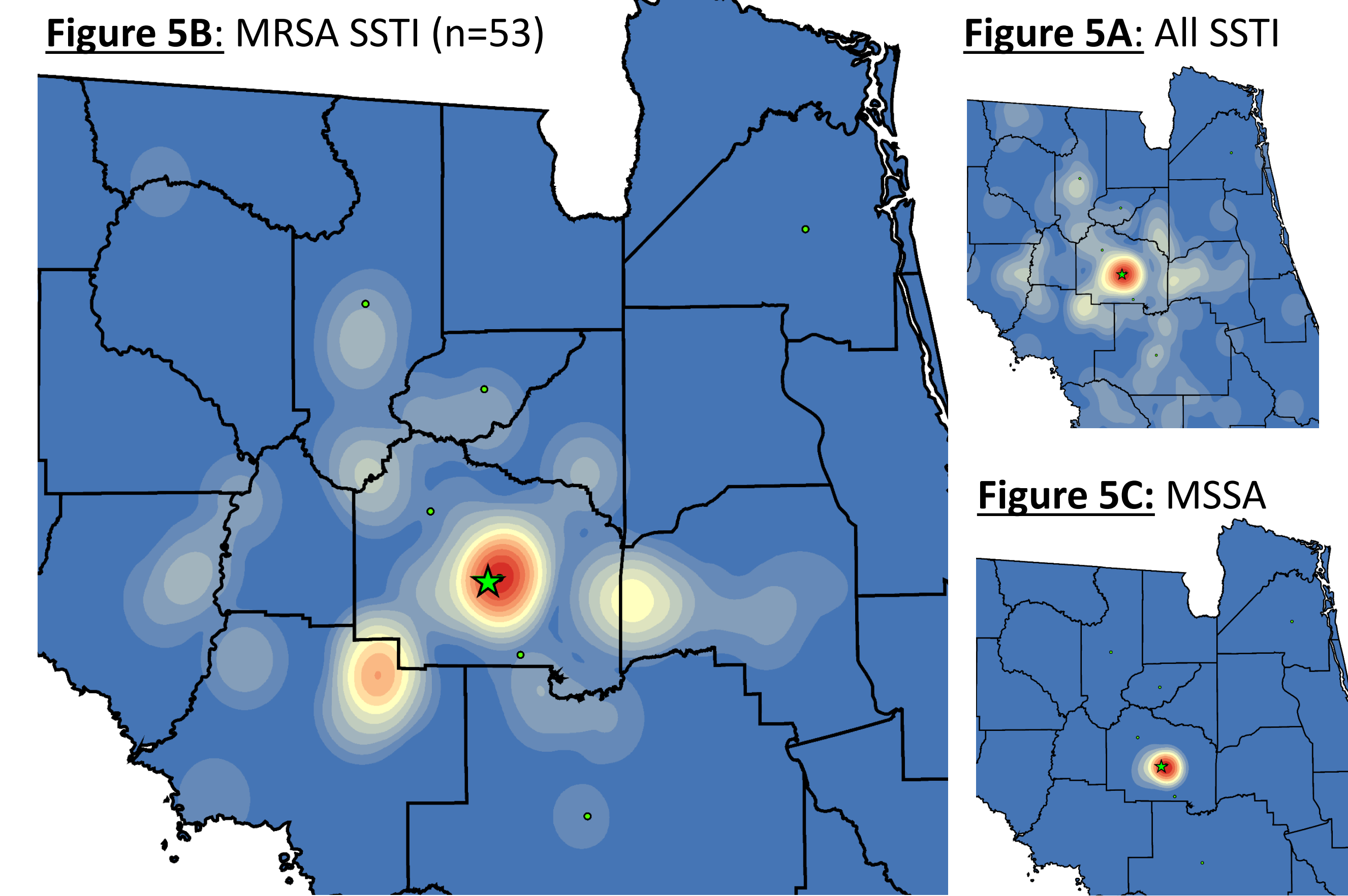


Table 1: Demographics and facotrs by community MRSA microbiology

	SSTI Culture		P Value
	MRSA (n=53)	Non-MRSA (n=129)	
Demographics:			
Average Age (± SD)	36.4 ± 21.0	37.8 ± 20.3	0.676
Male	31 (58.5)	66 (51.2)	0.415
Female	22 (41.5)	63 (48.8)	
Black	10 (18.9)	31 (24.0)	0.670
White	39 (73.6)	83 (64.3)	
Other	4 (7.5)	15 (11.7)	
Social History:			
Livestock Exposure	14 (26.4)	14 (10.9)	0.009
Rural	14 (26.4)	17 (13.1)	0.049
Urban	39 (73.6)	112 (86.9)	

Figure 5: Kernel density models illustrating spatial patterns of (A) all eligible subjects (B) MRSA SSTI (C) MSSA SSTI incidence by self-reported residence.



CONCLUSION

- MRSA transmission in hospital settings was introduced from multiple epidemiologically unrecognized MRSA species with ancestral origins in the community settings.
- The non-uniform geographic distribution, particularly high incidence in rural areas, suggest lifestyle and behavioral patterns may influence transmission in community settings more than genotypic factors.
- Nosocomial MRSA outbreak prevention strategies should target unique aspects of the community rather than solely on the hospital